



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Integrated Bachelor's-Master's Program

End-Semester Examination-Spring 2025

Subject: Electrodynamics and Electronics
Full marks: 50

Subject Code(s): PHS 2202
Time allotted: 3 hrs

Section A

Attempt any three questions

1. (a) Determine the scalar potential $\Phi(t, \vec{r})$ of a time varying charge distribution with charge density $\rho(t, \vec{r})$, from the finiteness of the speed of light (denoted by c). Hence show that the scalar potential satisfies the following equation: $-c^{-2}(\partial^2 \Phi / \partial t^2) + \nabla^2 \Phi = -\rho(t, \vec{r}) / \epsilon_0$. (7 marks)
(b) Given the above result, or, otherwise, find out the scalar potential due to a point charge q , moving with a velocity $\vec{v}$ and acceleration $\vec{a}$. (3 marks)

2. (a) Determine the electric and magnetic fields, the charge and the current distributions due to the following scalar and vector potentials:

$$\Phi(t, \vec{r}) = 0 ; \quad \vec{A}(t, \vec{r}) = -\frac{1}{4\pi\epsilon_0} \frac{qt}{r^2} \hat{r} .$$

- (b) Given the above charge and current distributions, can you suggest some other choices of $\Phi(t, \vec{r})$ and $\vec{A}(t, \vec{r})$, which can generate the same electric and magnetic fields? (3 marks)
(c) Find out the gauge transformation, connecting the scalar and the vector potential given in (a), with those derived in (b). (3 marks)
3. (a) Show that free charges inside a conductor dissipates away quickly. Hence write down the modified Maxwell's equations. (3 marks)
(b) Show that during the propagation of an EM wave through a conductor, the electric and magnetic fields experience a phase difference among them. Determine this phase difference and show that the phase difference becomes small for good conductors. (7 marks)
4. Consider a current carrying loop moving through a magnetic field, so that the magnetic flux passing through the loop changes with time. As a consequence a current flows through the loop. Show that, the emf generated in the loop is equal to the negative of the rate of change of magnetic flux. (5 marks)

- (b) Take a circular wheel of radius R , whose perimeter carries a line charge density λ . The wheel is placed in the (x, y) plane, with the origin of the wheel located at $x = 0 = y$. A uniform magnetic field $B_0 \hat{z}$ exists at the central region of the wheel, extending upto a radius $r_0 < R$. If the field is now turned off, the wheel starts to rotate. Determine the direction of rotation and angular velocity of the wheel. (5 marks)
5. (a) Determine the symmetry transformations between electric and magnetic fields that keeps the vacuum Maxwell's equations invariant. (3 marks)
- (b) Using the Maxwell's equations, show that,

$$\frac{\partial(u_{EM} + u_{Mech})}{\partial t} + \nabla \cdot \vec{S} = 0 ,$$

where, $\vec{S}$ is the Poynting vector, u_{EM} is the energy density of electromagnetic field and u_{mech} is the mechanical energy density of the charges and currents. (7 marks)

Section B

Attempt any four questions

1. Using the Thevenin equivalent circuit for Fig. 1 (a), calculate the values for I_L and V_L across the load R_L . (5 marks)

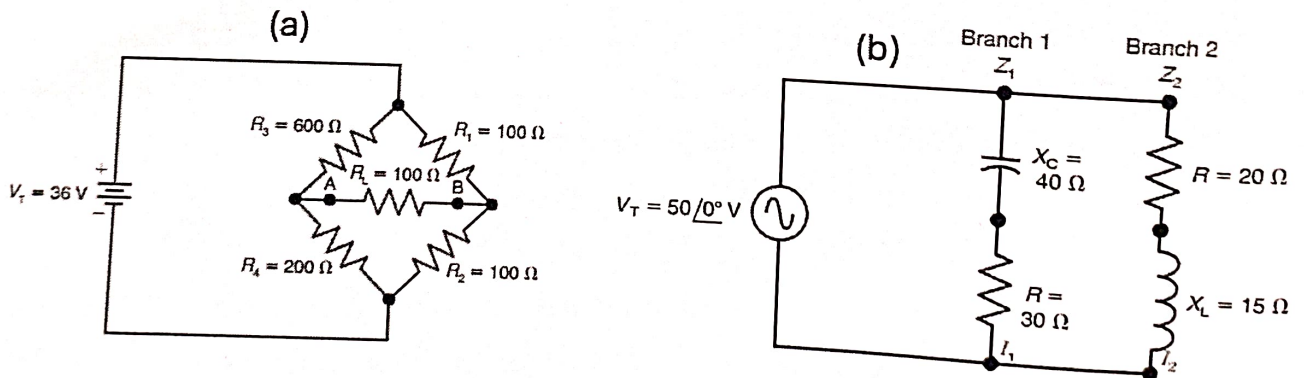


Figure 1: Electrical Circuit

2. Find the currents I_1 and I_2 in polar form ($|x| \angle \phi$) for the circuit given in Fig. 1 (b). Here, X_C and X_L are the magnitudes of capacitive and inductive reactances, respectively. (5 marks)
3. Consider an n-p-n bipolar junction transistor in CE mode. The transistor draws a constant base current of $I_B = 30 \mu A$. For a change in the collector current from 3.5 mA to 3.7 mA, the corresponding collector-emitter voltage changes from 7.5 V to 12.5 V. Calculate the output resistance and the β at $V_{CE} = 12.5$. (5 marks)

4. A CE amplifier has hybrid parameters $h_{ie} = 1500 \Omega$, $h_{fe} = 60$ and $h_{oe} = 12.5 \times 10^{-6} \Omega^{-1}$. The resistance to load is $5 \text{ k}\Omega$. Find the values of (i) current gain and (ii) voltage gain. You can assume the input impedance $Z_{in} \sim h_{ie}$. (5 marks)
5. Determine the output voltage for the circuit shown below. (5 marks)

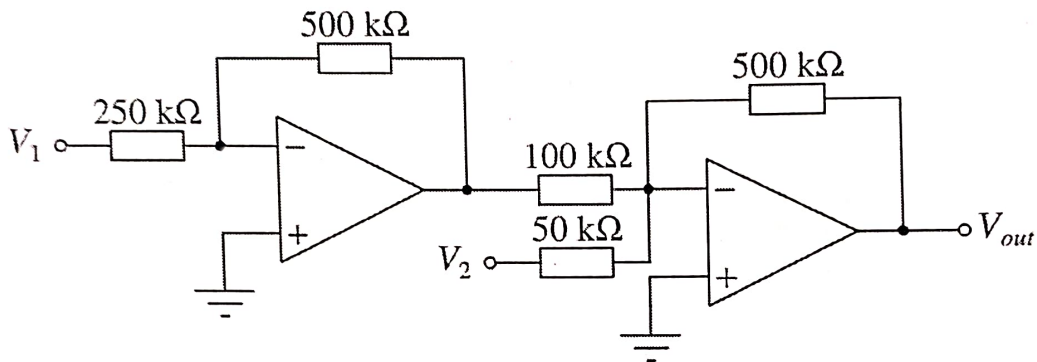


Figure 2: Amplifier Circuit

6. If $v_1(t) = 10\cos(2t) \text{ mV}$ and $v_2(t) = 0.5t \text{ mV}$ below, find $v_o(t)$ for $t > 0$. Assume that the voltage across the capacitor is zero at $t = 0$. This is a *summing integrator* circuit. (5 marks)

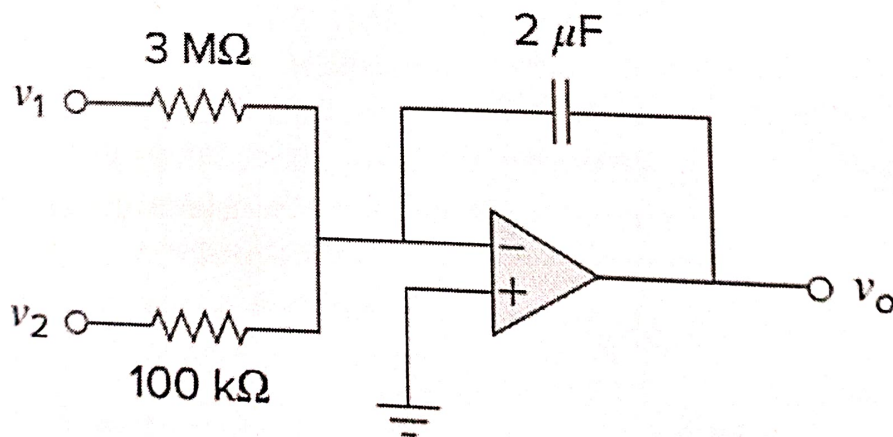


Figure 3: Integrator circuit



Indian Association for the Cultivation of Science
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Integrated Bachelor's-Master's Program

End-Semester Examination-Spring 2025

Subject: Classical Mechanics and Special Relativity
Full marks: 50

Subject Code(s): PHS 2201
Time allotted: 3 hr

1. (a) Define spacelike, timelike and null intervals. Show them in a light cone diagram. (5 marks)
(b) Using a light cone diagram, show that traveling faster than the speed of light may lead to 'causality' violation with respect to an event. (5 marks)
2. (a) Two inertial frames are in relative motion, with relative speed v along the x -axis. Find the relation between the components of velocity of a particle in the two frames following the laws of special relativity. (6 marks)
(b) If the speed of two spacecrafts A and B with respect to Earth's frame are $0.5c$ and $0.9c$, respectively, what is the speed of B as observed by A? (4 marks)
moving towards each other
3. (a) Explain the length contraction and time dilation using Lorentz transformation between two inertial frames in relative speed v along the x axis. (6 marks)
(b) A meter stick moving with respect to an observer appears 500 mm long to the observer. What is the relative speed if the meter stick is kept parallel to its direction of motion? How long does it take to pass her? (4 marks)
4. (a) Show that, if a particle of mass M decays into two particles of individual masses m_1 and m_2 , then we have $M \geq m_1 + m_2$. (4 marks)
(b) If in the problem above, $M = 500$ MeV, $m_1 = 100$ MeV and $m_2 = 200$ MeV, find the momenta p and the energy E_1 and E_2 of the daughter particles in the rest frame of M . Derive the necessary expressions for the above calculations. (6 marks)
5. (a) Find the number of degrees of freedom of a rigid body of N particles, where $N \geq 3$. What are Euler angles? (3 marks)
(b) Show that the rotational kinetic energy of a rigid body is $T_R = (1/2)\vec{\omega} \cdot \vec{I} \cdot \vec{\omega}$, where symbols have their usual meaning. (4 marks)
(c) Consider a cube of side a . Find the moment of inertia tensor of the cube, with the (x, y, z) axes directed along the edges of the cube. (3 marks)



INDIAN ASSOCIATION FOR THE CULTIVATION OF SCIENCE
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Master's/Integrated Master's-PhD Program/Integrated
Bachelor's-Master's Program/PhD Course
End-Semester Examination-Spring 2025

Subject: Quantum Mechanics
Full Marks: 50

Subject Code (s): PHS 2203
Time Allotted: 3 h

Attempt any five questions

1. Assume a particle moving in potential, $V = \frac{1}{2}kx^2$.
(a) Write down the Hamiltonian, H , using ladder operators. (2)
(b) Calculate the uncertainty product $\Delta x \Delta p$ for the above simple harmonic oscillator in the n -th state. Explain the physical interpretation of the increase of the uncertainty product $\Delta x \Delta p$ with quantum number n . (4+1)
(c) Calculate $[H, [p, x]] + [p, [x, H]]$. (3)

2. A particle in a harmonic oscillator potential is initially in the state,

$$\psi(x, 0) = A [2\psi_0(x) + 5\psi_1(x)]$$

where $\psi_n(x)$ are the energy eigenfunctions of the harmonic oscillator.

- (a) Derive the time-dependent wave function $\psi(x, t)$ and find the probability density $|\psi(x, t)|^2$. Explain why the probability density depends on time. How does this time-dependence relate to the "conservation of probability"? (3+2+2)
(b) What are the possible energy values, and their probabilities? (3)
3. The hydrogen atom is described by the Coulomb potential,

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

where e is the electron charge and ϵ_0 is the vacuum permittivity.

- (a) Write down the time-independent Schrödinger equation for an electron in spherical coordinates and separate the equation into radial and angular parts. (3)
(b) From the radial equation, find the allowed energy eigenvalues, E_n . (4)
(c) Consider a particle is in state of linear combination of spherical harmonics, $\psi(\theta, \phi) = \sqrt{\frac{2}{13}}Y_{3,3} + \sqrt{\frac{3}{13}}Y_{3,-2} + \sqrt{\frac{3}{13}}Y_{3,0} + \sqrt{\frac{3}{13}}Y_{3,2} + \sqrt{\frac{2}{13}}Y_{3,-3}$. Calculate the expectation values of L_z and L^2 operators. (3)



Indian Association for the Cultivation of Science
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Integrated Bachelor's-Master's Program

End-Semester (UG-IV) Examination- May 2025

Subject: Thermodynamics and Statistical Mechanics Subject Code: PHS

2204

Full marks: 50

Time allotted: 3 hrs

1 Questions

1. Evolution of entropy: the normalized ensemble density is a probability in the phase space $\Gamma(p, q)$. This probability has an associated entropy $S(t) = - \int d\Gamma \rho(\Gamma, t) \ln \rho(\Gamma, t)$.

- (a) Show that if $\rho(\Gamma, t)$ satisfies Liouville's equation for a Hamiltonian H , $\frac{dS}{dt} = 0$.

[10]

(Hint: Use $\frac{\partial \rho}{\partial t} = -\{\rho, H\}$ inside the integral and integrate it by parts.)

- (b) Using the method of Lagrange multipliers, find the function $\rho_{max}(\Gamma)$ that maximizes the functional S , subject to the following constraints:

Fixed average energy, $\langle H \rangle = E = \int d\Gamma \rho(\Gamma) H$ and

Normalization of ρ , $\int d\Gamma \rho(\Gamma) = 1$.

(Hint: Use Lagrange multipliers to maximize the entropy functional. The answer will be of the form $\rho_{max} = Ce^{-\beta H}$ where β is the Lagrange multiplier corresponding to fixed energy constraint.)

Also show that $\frac{\partial \rho_{max}}{\partial t} = 0$. [3+2]

2. consider rod-shaped molecules with moment of inertia I , and a dipole moment μ . The Hamiltonian of the system corresponding to rotational degrees of freedom is given by

$$H_{rod} = \frac{1}{2I} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \right) - \mu E \cos \theta \quad (1)$$

where E is an external electric field. $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ are the polar and azimuthal angles, and p_θ, p_ϕ are their conjugate momenta.

- (a) Compute the classical partition function corresponding to the rotational degrees of freedom of each dipole. [10]

(b) Obtain the mean polarization $P = \langle \mu \cos \theta \rangle$ and mean energy from the Partition function. [5]

(Hint: You need to show $Z_{rot} \propto \frac{\sinh(\mu\beta E)}{\mu\beta E}$)

3. Consider a classical system of N non interacting diatomic molecules enclosed in a three dimensional box of volume V at temperature T . The Hamiltonian for a single molecule is taken to be

$$H(p_1, p_2, q_1, q_2) = \frac{1}{2m}(p_1^2 + p_2^2) + \frac{1}{2}K|\vec{q}_1 - \vec{q}_2|^2 \quad (2)$$

where $p_{1,2}, q_{1,2}$ are the momenta and position coordinates of two atoms in the molecule.

Write down the partition function for N molecules and obtain the Free energy from it. [10]

Hint: Use change of variables $\vec{x} \equiv \frac{\vec{q}_1 - \vec{q}_2}{2}, \vec{y} \equiv \frac{\vec{q}_1 + \vec{q}_2}{2}$, where the integration range of $|\vec{x}|$ can be approximated by $(0, \infty)$. You don't have to keep track of numerical constants in the integration, write the final expression in terms of solid angle Ω_n (for n dimensional sphere).

4. Consider a system of N non-interacting, identical and distinguishable particles, each of which can occupy any two of the energy states with energies ϵ and $-\epsilon$. Compute the entropy of the system in microcanonical ensemble and in canonical ensemble as a function of mean energy per particle x . Show that in the thermodynamic limit, both coincides. [10]

(Hint: $x \equiv \frac{U}{N\epsilon}$. For microcanonical ensemble, $U = n_+\epsilon - n_-\epsilon$ and $N = n_+ + n_-$. You should write $n_{\pm}$ in terms of $x \equiv \frac{U}{N\epsilon}$ to compute the entropy. For canonical partition function, use $Z_N = Z_1^N$, where Z_1 is the single particle partition function.)



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
End-Semester Examination-Spring 2025

*Subject: Introduction to Mathematical Methods
in Physics (SECC)
Full Marks: 50*

*Subject Code(s): PHS 2279
Time Allotted: 3 h*

1. Answer any 5 from the following:

(5 x 2=10)

- a. Show if the series diverges or converges: $\sum_{n=1}^{\infty} (-1)^n \frac{n!}{\pi^n}$
- b. Show limit doesn't exist for : $\lim_{(x,y) \rightarrow (0,0)} \frac{(x+y)^2}{x^2+y^2}$
- c. At what points on the plane is the function continuous : $g(x, y) = \sin \frac{1}{xy}$
- d. A thin elastic membrane is stretched over a square frame . The height z of the membrane over a point (x, y) on the square is given by $z = x^2 - y^2 + \sin(xy)$. Find all the critical points and determine the saddle point.
- e. Find the directional derivative of $\varphi = x^2y + xz$ at $(1, 2, -1)$ in the direction $\vec{A} = 2\hat{i} - 2\hat{j} + \hat{k}$.
- f. Prove that cylindrical co-ordinate system is orthogonal.
- g. Rotate a point $(2, 3)$ counterclockwise by $\theta = 45$ degree about origin using rotation matrix.

Answer any 4 from the following:

(4 x 10 = 40)

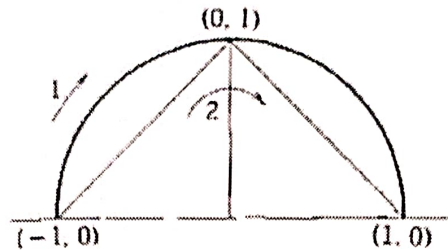
2. a. If $x^y = y^x$, find $\frac{dy}{dx}$.

b. Find the volume in the first octant bounded by cylinder $x^2 + y^2 = 2$ and the planes $z = x + y, y = x, z = 0$ and $x = 0$.

c. Find the power series representation of $\ln(1 + x)$.

(3+4+3)

3. a. Find the value of $I = \int \frac{xdy - ydx}{x^2 + y^2}$ along two of the paths indicated in the figure from $(-1,0)$ to $(1,0)$.



- b. Solve the differential equation: $x^4 \frac{dy}{dx} + x^3 y = -\sec(xy)$
- c. Solve the simultaneous equations by Cramer's rule: $x + 2y + 3z = 14$; $2x - y + z = 5$; $3x + y + 2z = 10$; (5+2+3)
4. a. The temperature in a rectangular plate bounded by $x=0$, $y=0$, $x=3$, $y=5$ is $T = xy^2 - x^2y + 100$. Find the hottest and coldest point of the plate.
- b. Radium decays to radon which decays to polonium. If at $t=0$, a sample is pure radium, how much radon does it contain at time t ? (5+5)
5. a. If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, evaluate $\iiint \nabla \cdot \vec{F} dV$, where V is the closed region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $2x + 2y + z = 4$.
- b. Test the convergence of series $\frac{\sqrt{2}-1}{3^3-1} + \frac{\sqrt{3}-1}{4^3-1} + \frac{\sqrt{4}-1}{5^3-1} + \dots$ (5+5)
6. a. Verify Stoke's theorem for $\vec{A} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is the boundary of the surface S .
- b. Find the general solution of $\frac{d^2y}{dx^2} - y = x \sin 3x + \cos x$ (6+4)
7. a. Find the general solution of $y'' - 2y' + 2y = x + e^x \cos x$
- b. Explain the physical significance of Taylor Series taking an example of $f(x) = e^x$
- c. Find the solution of $y'' = y$, which passes through the origin and through the point $(\ln 2, \frac{3}{4})$. (4+3+3)



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Integrated Bachelor's-Master's Program

Mid-Semester Examination-Spring 2025

Subject: Electrodynamics and Electronics
Full marks: 25

Subject Code(s): PHS 2202
Time allotted: 2 hr

Attempt any five questions

- ✓1. State and prove uniqueness theorem in electrostatics. (5 marks)
2. A dipole $\vec{\mu} = q\vec{\ell}$ is kept in an external electric field $\vec{E}$. Show that the potential energy of the dipole is given by $-\vec{\mu} \cdot \vec{E}$. (5 marks)
3. Consider a conducting sphere of radius R , which is grounded and a charge q is placed at a distance D from the center of the sphere. Determine the magnitude and sign of the image charge, as well as its location. (5 marks)
4. Given that the vector potential due to a current distribution is given as,

$$\vec{A}(\vec{r}) = \mu_0 \int \frac{\vec{J}(\vec{r}')}{|\vec{r} - \vec{r}'|} dV',$$

where, $\vec{r}'$ is the source coordinate and $\vec{r}$ is the field coordinate from a suitably chosen origin and $\vec{J}$ is the current density. Using the above, determine the magnetic field and hence show that

$$\vec{\nabla} \cdot \vec{B} = 0.$$

Discuss the implications of this relation.

- ✓5. Find the magnetic field at a perpendicular distance r from an infinitely long current carrying wire, carrying current I . (5 marks)
6. (a) In the common emitter configuration of an NPN bipolar junction transistor, the base resistance $R_B = 100 \text{ k}\Omega$ and the collector resistance $R_C = 1 \text{ k}\Omega$. Draw a schematic of the circuit. For a base current of $50 \text{ }\mu\text{A}$, a voltage drop of 5 V is recorded across R_C . Calculate the α of the transistor. (2 marks)
- ✓(b) If a certain transistor has a maximum power output of 1.2 W , what will be the value of collector bias voltage V_{CC} for a collector current of 50 mA ? (1 mark)
- (c) Show that for a bipolar junction transistor,

$$I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO}$$

Here I_{CBO} is the collector base current, when the emitter is open.

(2 marks)



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
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Mid-Semester Examination-Spring 2025

Subject: *Introduction to Mathematical Methods*
in Physics (SECC)
Full Marks: 25

Subject Code(s): *PHS 2279*
Time Allotted: 2 h

1. Find the limit if it exists or show that the limit doesn't exist. [2]

(i) $\lim_{(x,y) \rightarrow (1,0)} \ln\left(\frac{1+y^2}{x^2+xy}\right)$

(ii) $\lim_{(x,y,z) \rightarrow (\pi, 0, \frac{1}{3})} e^{y^2} \tan(xz)$

2. Determine the sets of points at which the function is continuous [2]

(i) $F(x, y) = \tan^{-1}((x+y)^{-2})$

(ii) $f(x, y) = \begin{cases} \frac{xy}{x^2+xy+y^2}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$

3. Verify that the function $u = 1/(\sqrt{x^2 + y^2 + z^2})$, is a solution of 3-dimensional Laplace equation. [3]

OR

Verify that the function $u = e^{-\alpha^2 k^2 t} \sin(kx)$ is a solution of heat conduction equation, where α is the thermal diffusivity of the material. [3]

4. If $f(x, y) = xe^y$, find the rate of change of f at point $P(2,0)$ in the direction from P to $Q(1/2, 2)$. In what direction does f have maximum rate of change? What is this maximum rate of change? [3]

OR

A function gives as $f(x, y) = x^2 + y^2$. The directional derivative of f at point $(1,2)$ are given in the direction of vectors $\vec{u} = 3\hat{i} + 4\hat{j}$ and $\vec{v} = -4\hat{i} + 3\hat{j}$ as $D_u = 10, D_v = 4$. Find the gradient $\nabla f(1,2)$. [3]

5. Use Lagrange multiplier to prove that triangle with maximum area that has a given perimeter p is equilateral. [2]



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Subject: Quantum Mechanics (PHS 2203)
Date: March 06, 2025

Full Marks: 25
Timing: 11:00 hrs to 13:00 hrs

Attempt any one of the following questions: 1 or 2.

1. (a) Show that eigenvalues of a Hermitian operator are real. (2)
(b) Find when the product of two hermitian operators is also hermitian. (1)
(c) Show that the momentum operator $\hat{p}$ is hermitian. (2)
2. For a unitary operation, $\hat{T}(a)\Psi(x) = \Psi(x + a)$, here $\hat{T}(a)$ is an unitary operator.
(a) Find the expression of $\hat{T}(a)$ in terms of momentum. Explain that the momentum is the generator of translation. (2)
(b) Show that for continuous translation symmetry (i.e. $[\hat{H}, \hat{T}] = 0$), the momentum remains conserved. (3)

Attempt any two of the following questions: 3, 4, or 5.

3. Assume a particle in a initial state, $\Psi(x, 0) = A[\Phi_1(x) + \Phi_2(x)]$. Here, Φ_1 and Φ_2 are the ground state and first excited state wave function of the particle in an infinite square-well potential of width a . Here, energy $E_n = n^2 E_1$.
After time t ,
(a) Write the wave function. (1)
(b) Calculate the probability of finding the particle in the first excited state. (3)
(c) Calculate the expectation value of x . (4)
(d) Find the expectation value of H . (2)
4. Consider a particle of mass m moving in a delta potential $V(x) = -\alpha\delta(x)$.
(a) Solve the Schrodinger equation to find the wave function of the particle of energy $E < 0$. Find the discontinuity in momentum, p at $x = 0$. (4)
(b) Show that the potential can support exactly one bound state regardless the value of α . (1)
(c) Determine the uncertainty in position and momentum (i.e. σ_x and σ_p). Show that they follow the Heisenberg uncertainty relation. (5)



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Integrated Bachelor's-Master's Program

Mid-Semester (UG-IV) Examination- Spring 2025

Subject: Thermodynamics and Statistical Mechanics Subject Code: PHS

2204

Full marks: 25

Time allotted: 2 hrs

1 Questions

1. A body of constant heat capacity C_P and at a temperature T_i , put in contact with a reservoir at a higher temperature T_f . The pressure remains constant while the body comes to equilibrium with the reservoir. Show that the entropy change of the universe is equal to

$$\Delta S_{univ} = C_P[x - \ln(1+x)]; \text{ where } x = \frac{T_i - T_f}{T_f}.$$

Also proof that the total entropy change of the universe is positive.

Hint: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$, when $x < 1$.

(5)

2. Starting from the mathematical formulation of second law (for reversible process) for any thermodynamic substances, proof the following identities:

$$TdS = C_V dT + T \left(\frac{\partial P}{\partial T} \right)_V dV, \quad (1)$$

$$TdS = C_P dT - T \left(\frac{\partial V}{\partial T} \right)_P dP. \quad (2)$$

Also from these, prove the following:

$$C_P - C_V = T \left(\frac{\partial V}{\partial T} \right)_P \left(\frac{\partial P}{\partial T} \right)_V$$

Hint: To prove (1), expand U in terms of T and V . To derive (2), expand U and V in terms of T and P . Then use Maxwell's equation for dS .

(10)

3. Prove that it is impossible for two reversible adiabatics (with two independent variables) to intersect. (Hint: Assume that they do intersect and complete the cycle

with an isothermal. Show that the performance of this cycle violates the second law.)

If there are more than two independent thermodynamic variables, why two adiabatics can intersect? What does it signify?

(5)

4. For a reversible isothermal process, what is the relation between Helmholtz Free energy and work done by the system? Using this relation and the Maxwell equation for Free energy, compute the free energy of a thermodynamic substance $F(V, T)$ where the following properties are given:

- At a constant temperature of T_0 the work done by it on expansion from V_0 to V is $W = RT_0 \log \frac{V}{V_0}$.
- The entropy is given by $S = R \frac{V}{V_0} \left(\frac{T}{T_0} \right)^3$.
- Here V_0, T_0 are constants.

(5)



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Integrated Bachelor's Master's Programme

Mid-Semester Examination-Spring 2025

Subject: Classical Mechanics and Special Relativity
Full marks: 25

Subject Code(s): PHS 2201
Time allotted: 2 hr

1. (a) Write the Lagrangian for a pendulum of mass m and length L , whose point of suspension moves up with an acceleration $a(t) = (1/2)at^2$. From this find the time period of the pendulum. (3 marks)
(b) Write the Lagrangian of a particle moving in a central force field described by a potential $V(r)$. From this find the conserved quantities. (2 marks)
2. (a) Show that if $L(q, \dot{q}, t)$ satisfies Lagrange's equation, then $L' = L + df(q, t)/dt$ also satisfies Lagrange's equation where $f(q, t)$ is an arbitrary well-behaved function of q and t . (3 marks)
(b) Show that for a coordinate transformation $r_i = r_i(q_1, q_2, \dots, q_n, t)$, we have, $(\partial \dot{r}_i / \partial \dot{q}_j) = (\partial r_i / \partial q_j)$. (2 marks)
3. (a) Using Hamilton's equation of motion, show that for any dynamical variable $A(q, p, t)$, $(dA/dt) = \{A, H\} + (\partial A / \partial t)$, where H is the Hamiltonian of the system. (3 marks)
(b) From the above, find (dp_x/dt) for a particle moving along x -axis, in a potential $V(x)$. (2 marks)
4. (a) From the definition of Hamiltonian $H = \sum_i \dot{q}_i (\partial L / \partial \dot{q}_i) - L$, find when H can be interpreted as the total energy of the system. (3 marks)
(b) Find the condition for H to be a conserved quantity. (2 marks)
5. (a) Two masses M_1 and M_2 are hanging over a pulley, where M_1 and M_2 are connected by an inextensible string of length L . Write the Lagrangian of the system and find the motion of M_1 as a function time. (3 marks)
(b) What is meant by cyclic coordinate? Find the cyclic coordinate for projectile motion of a mass m near the surface of Earth. (2 marks)

UG4 (Electrodynamics - PHS 2202)

Class Test 1

Full marks : 25 Time : 1 hr 20 mins

[Answer question number (1) and any four among the rest of the questions. Each question carries 5 marks.]

1. Answer all the questions. Each question carries 1 marks.

a) What are the boundary conditions on the interface of two different layers of media?

b) Suppose you have a very thin layer of surface and rest of the the space is free from any medium. How do you get the boundary conditions for free space from the boundary conditions you have found in question number (1.a)?

c) Using the mathematical expression of electric dipole moment $\vec{p}$, find the electric field at a point $\vec{x}$ due to the dipole situated at $\vec{x}_0$ in a coordinate independent form.

d) What is electric polarization $\vec{P}$ and electric displacement $\vec{D}$?

e) If you specify the potential on the boundary of a surface, then show that the Green's function and hence argue that the potential everywhere can be found uniquely.

2. A cubical box of side length a has been made using five metal plates (welded together and grounded) and a separate sheet of metal, insulated from the others and is maintained at a constant potential V_0 .

a) Write down the boundary conditions.

[1]

b) Write down the differential equation to solve the potential, use the separation of variables and find the potential inside the cubical box.

[4]

3. Two semi-infinite grounded conducting planes ($x = 0$ and $y = 0$) meet at right angles. A point charge q is at (a, b) .

a) Calculate the potential in the region where the charge is located using the method of images.

[3]

b) What is the force on q ?

[1]

c) How much work did it take to bring q from infinity?

[1]

4. Charge density $\sigma(\theta) = a \sin 5\theta$ (where a is constant, θ is the azimuthal angle of a cylinder) is glued over the surface of an infinite cylinder of radius R whose principal axis is along the Z-axis.

(a) Find the potential inside the cylinder.

[2.5]

(b) Find the potential outside the cylinder.

[2.5]

5. A charge distribution has multipole moments $q, \vec{p}, Q_{ij}, \dots$ for one set of coordinate axes and moments $q', \vec{p}', Q'_{ij}, \dots$ for another set of coordinate axes whose axes are parallel to the first set but origin is shifted to a point $O' = (a, b, c)$ relative to the first.

a) Determine explicitly the connections between the monopole, dipole and quadrupole moments in the two coordinate frames. [3]

b) If $q \neq 0$, can O' be found such that $\vec{p}' = 0$? [1]

c) If $\vec{p} \neq 0$, can O' be found so that $Q'_{ij} = 0$? Is it irrespective of the choice of q ? [1]

6. A localized charge density $\rho(x, y, z)$ is placed in an external electrostatic field described by a potential $\Phi_0(x, y, z)$. The external potential varies slowly in space over the region where the charge density is different from zero.

a) Calculate the total force $\vec{F}$ acting on the charge distribution as an expansion in multipole moments up to quadrupole moments.

$$\phi = -\nabla \cdot \vec{E} \quad [2]$$

b) Calculate the energy W and find its connection with $\vec{F}$. [1]

c) Repeat the calculation to find the total torque $\vec{N}$ on the particle. [2]

7. A very long, right circular, cylindrical shell of dielectric constant $\frac{\epsilon}{\epsilon_0}$ with inner radius a and outer radius b is placed in a uniform electric field E_0 with its axis perpendicular to the field. The medium inside and outside the cylinder has a dielectric constant of unity.

a) Determine the potential and electric field in the three regions. [2+2=4]

~~a) Calculate the total force $\vec{F}$ acting on the charge distribution as an expansion in multipole moments up to quadrupole moments.~~

b) Sketch the field lines considering $b \approx 2a$ [1]

8. Two concentric conducting spheres of inner and outer radii a and b , respectively, carry charges $\pm Q$. The empty space between the spheres is half-filled vertically by a hemispherical shell of dielectric constant $\frac{\epsilon}{\epsilon_0}$.

a) Find the electric field everywhere between the spheres. [4]

b) Calculate the surface-charge distribution on the inner sphere. [1]

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Class Test 2

Full marks : 20 Time : 1 hr 20 mins

[Answer any two questions. Each question carries 10 marks.]

1. a) A cylindrical conductor of radius a has a hole of radius b bored parallel to, and centered a distance d from, the cylinder axis ($d + b \leq a$). The current density is uniform throughout the remaining metal of the cylinder and is parallel to the axis. Use Ampère's law and principle of linear superposition to find the magnitude and the direction of the magnetic-flux density in the hole.

[5]

b) i) A circuit consists of a long thin conducting shell of radius a and a parallel return wire of radius b on axis inside. If the current is assumed distributed uniformly throughout the cross-section of the wire, calculate the self-inductance per unit length.

[3]

ii) What is the self-inductance if the inner conductor is a thin hollow tube?

[2]

2.) A circular current loop of radius a carrying a current I lies in the x - y plane with its center at the origin.

(a) Show that the only nonvanishing component of the vector potential is

$$A_\phi(\rho, z) = \frac{\mu_0 I a}{\pi} \int_0^\infty dk \cos kz I_1(k\rho_<) K_1(k\rho_>),$$

where $\rho_<$ ($\rho_>$) is the smaller (larger) of a and ρ .

[3]

(b) Show that an alternative expression for A_ϕ is

$$A_\phi(\rho, z) = \frac{\mu_0 I a}{2} \int_0^\infty dk e^{-k|z|} J_1(ka) J_1(k\rho).$$

[3]

(c) Write down integral expressions for the components of magnetic induction using the expressions of (a) and (b). Evaluate explicitly the components of $\mathbf{B}$ on the z axis by performing the necessary integrations.

[4]

3) A circular loop of wire carrying a current I is located with its center at the origin of coordinates and the normal to its plane having spherical angles θ_0, ϕ_0 . There is an applied magnetic field,

$$B_x = B_0(1 + \beta y), \quad B_y = B_0(1 + \beta x).$$

(a) Calculate the force acting on the loop without making any approximations.

[5]

(b) Calculate the torque in lowest order. Can you deduce anything about the higher order contributions? Do they vanish for the circular loop? What about for other shapes?

[5]

4) Two concentric circular loops of radii a, b and currents I, I' respectively ($b < a$), have an angle α between their planes. Show that the torque on one of the loops is about the line of intersection of the two planes containing the loops and has the magnitude:

$$N = \frac{\mu_0 \pi I I' b^2}{2a} \sum_{n=0}^{\infty} \frac{(n+1)}{(2n+1)} \left[\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n+2)\Gamma(\frac{3}{2})} \right]^2 \left(\frac{b}{a} \right)^{2n} P_{2n+1}^1(\cos \alpha),$$

where $P_{2n+1}^1(\cos \alpha)$ is an associated Legendre polynomial. Determine the sense of the torque for α an acute angle and the currents in the same (opposite) directions.

[5]

b) Show that in general a long, straight bar of uniform cross-sectional area A with uniform lengthwise magnetization M , when placed with its flat end against an infinitely permeable flat surface, adheres with a force given approximately by

$$F \approx \frac{\mu_0}{2} A M^2$$

[5]

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Class Test 3

Full marks : 20 Time : 1 hr 05 mins

[Answer any two questions. Each question carries 10 marks.]

1. a) Starting from Maxwell's Equations in the absence of any medium, find the Helmholtz wave equations from where you can find $\mathbf{E}(\vec{x}, t)$ and $\mathbf{B}(\vec{x}, t)$ in terms of plane wave fields. Write down the expressions for $\mathbf{E}(\vec{x}, t)$ and $\mathbf{B}(\vec{x}, t)$ and define phase velocity.

[1 + 1 = 2]

b) How can you prove that an electromagnetic wave is a transverse wave, given $\mathbf{E}(\vec{x}, t)$, $\mathbf{B}(\vec{x}, t)$, and wave vector $\vec{k} = k\hat{n}$. Also, show that you can find $\mathbf{B}(\vec{x}, t)$ from $\mathbf{E}(\vec{x}, t)$ and $\vec{k}$.

[1 + 1 = 2]

c) Show that for real $\hat{n}$; $\mathbf{E}(\vec{x}, t)$ and $\mathbf{B}(\vec{x}, t)$ have same phase while for complex valued $\hat{n}$ (Explain how this can be); the wave possesses exponential growth or decay term.

[1 + 1 = 2]

d) In the case of waves with no phase difference between $\mathbf{E}(\vec{x}, t)$ and $\mathbf{B}(\vec{x}, t)$, show how one can define linearly polarized, elliptically polarized wave by defining polarization vectors and writing most general homogeneous plane wave propagating in the direction $\vec{k} = k\hat{n}$.

[1 + 1 = 2]

e) How can one get a circularly polarized wave from an elliptically polarized wave? Write down the conditions under which we get left circularly polarized wave and right circularly polarized wave and justify your answer. For which limiting condition do we get back linearly polarized wave back from circularly polarized wave.

[1 + 1 = 2]

2. Consider there are two media separated by the $z = 0$ plane, where the media below and above have permittivities and permeabilities $\mu_1, \epsilon_1, \mu_2, \epsilon_2$ respectively. A plane wave with wave vector $\vec{k}$ and wave number ω travels from the first medium and is incident on the second medium, the reflected and refracted waves have wave vector and wave number as $\vec{k}_1, \omega_1$ and $\vec{k}_2, \omega_2$ respectively.

a) Write down electric and magnetic fields for all three waves, along with (i) phase factors at $z = 0$ and their relations, and (ii) the proper boundary conditions at $z = 0$.

[1 + 1 + 1 = 3]

b) Prove the laws of reflection and the laws of refraction.

[1]

c) Now, for simplicity, consider a linearly polarized wave. Two simple conditions are available from the boundary conditions: (i) the electric field is perpendicular to the plane of incidence, and (ii) the electric field is parallel to the plane of incidence. In both cases, show how the amplitudes of incident, reflected, refracted wave are connected.

$$[2 + 2 = 4]$$

d) Show that for total internal reflection, the refracted wave propagates only parallel to the surface and is attenuated exponentially beyond the interface. Also, show that no energy flows through the surface by calculating the Poynting vector.

$$[2]$$

3. a) Consider a point P moves with a velocity $\vec{u}'$ with respect to an inertial frame K' which itself moves with a velocity $\vec{u}$ with respect to another inertial frame K . If the point P is moving with a $\vec{v}$ with respect to the inertial frame K , find v in terms of u and u' and write γ_v in terms of γ_u and $\gamma_{u'}$.

$$[2 + 2 = 4]$$

b) Starting from the definition of the second rank antisymmetric field strength tensor $F^{\alpha\beta}(x)$ and its transformation under coordinate transformation, find how do components of electric and magnetic field change under Lorentz transformation.

$$[3 + 3 = 6]$$